Moving sentinel world: communication contrast depends on field of view

The hidden-state/action relation for this extension is summarized in the categorical loop schematic fig. 8; the results below remain the executed moving-world comparisons,

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Moving sentinel world (part 2)

not a claim that the schematic's full loop is present in every flat belief-sharing study.

Moving sentinel world (part 3)

The static sentinel world lets every agent observe the same shared latent, so belief sharing is a refinement rather than a requirement. To stress the *necessity* of communication we add movement and **disjoint** fields of view.

Moving sentinel world (part 4)

The world is a linear grid of 4 cells holding a single binary threat — left half (state 0) or right half (state 1). The 2 sentinels start at evenly tiled positions and each observe a half-open window of cells, so in the default setup agent 0 watches the left half and agent 1 the right half: their views do not overlap.

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Moving sentinel world (part 5)

Each agent's likelihood is a confident, signed presence reading for the half it can see, and three control paths (stay / left / right) let it reposition. The expected-free-energy policy scores each candidate move by the expected posterior entropy after one observation and takes the most information-seeking step.

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Moving sentinel world (part 6)

We run 960 trials of 6 steps each under three conditions: *isolated* (random moves, no sharing), *communicating* (random moves plus a log-linear-pool consensus each step), and *EFE-guided* (information-seeking moves plus the same sharing).

Moving sentinel world (part 7)

The measured consensus accuracies are 0.999 (isolated), 0.977 (communicating), and 0.978 (EFE-guided), with a communicating free-energy gap of -0.528 nats relative to the isolated baseline (negative:

Moving sentinel world (part 8)

no free-energy advantage over isolated in this binary-complement regime of logically complete half-views) (fig. 29).

Moving sentinel world (part 9)

Across 128 independent seeds the EFE-guided accuracy is 0.983 (95 % CI 0.982–0.984), the communicating (random-moves + sharing) accuracy is 0.982 (95 % CI 0.982–0.983), and the isolated accuracy is 0.999 (95 % CI 0.999–0.999).

Moving sentinel world (part 10)

In this binary-complement regime the isolated condition is in fact significantly *higher* on accuracy than the EFE-guided sharing condition — their 95 % intervals do not overlap — and the EFE-vs-isolated accuracy contrast yields Wilcoxon signed-rank $p = 9.27 \times 10^{-23}$ (significant; isolated higher),

Moving sentinel world (part 11)

effect size $r = 1.000$ (large). Sharing is therefore not merely unnecessary in this regime; it costs a small but reliable amount of accuracy.

Moving sentinel world (part 12)

Nor does sharing lower free energy here: the EFE free-energy gap (isolated surprise minus the EFE-guided condition's surprise) is -0.368 nats (95 % CI -0.384—0.353) — negative, so the pooled consensus is slightly *more* surprised by the true state than the isolated baseline, because a single agent's view already suffices.

Moving sentinel world (part 13)

The accuracy case for *necessity* is therefore made only in the larger-state-space disjoint-FOV extension below, not by these binary-complement numbers.

Moving sentinel world (part 14)

We report these numbers as measured, not assumed. The binary world carries a logical complement: ruling out one's own half implies the other, so a single agent's "not detected" still carries information about the global state, and an isolated agent is not strictly blind.

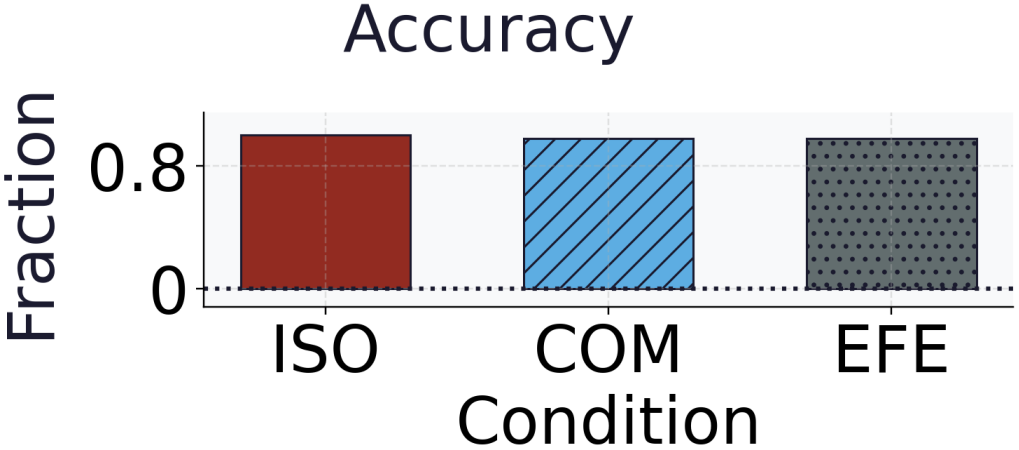
Moving sentinel world (part 15)

By design, the intended *cannot-decide-alone* regime is the high-noise, few-step corner where one sensor's evidence cannot overcome the flat prior; there belief sharing is meant to fuse the two complementary views into a decisive consensus. That regime is not separately measured here — the construction,

Moving sentinel world (part 16)

the three actions, the EFE rule, and the exact condition protocol are detailed in the supplement (sec. 17.8).

Moving sentinel world (part 17)

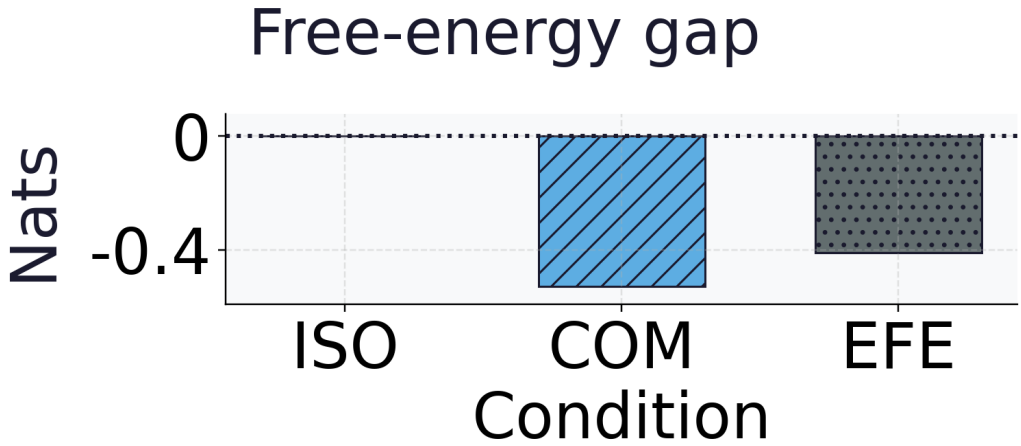


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Moving sentinel world (part 18)

Accuracy (fraction): ISO +0.999; COM +0.9771; EFE +0.9781.

Moving sentinel world (part 19)

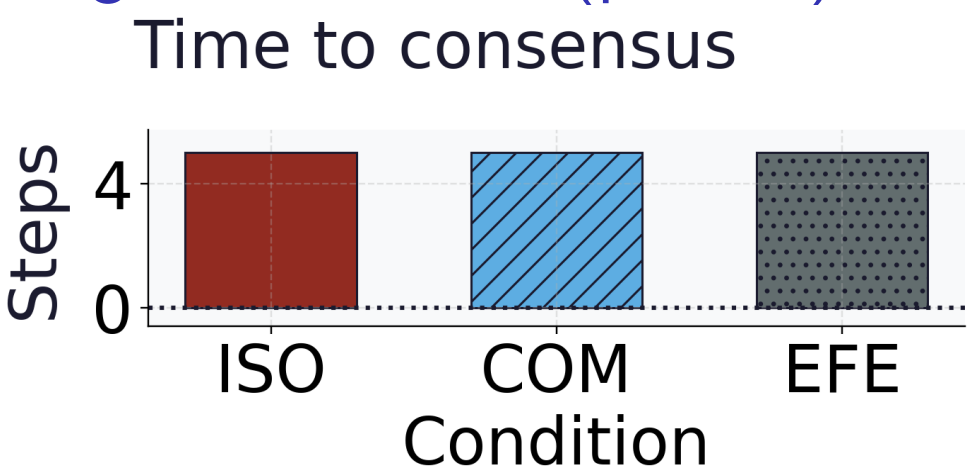


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Moving sentinel world (part 20)

Free-energy gap (nats): ISO +0; COM -0.5285; EFE -0.4099.

Moving sentinel world (part 21)



Moving sentinel world (part 22)

Time to consensus (steps): ISO +5.002; COM +5; EFE +5.001.

Moving sentinel world (part 23)

Accuracy gains over isolated: COM -0.022; EFE -0.021.

Moving sentinel world (part 24)

ISO / plain: isolated; COM / forward hatch: communicating; EFE / dots: EFE-guided. Different metric units have separate axes.

Moving sentinel world (part 25)

Declared moving-world summary. No interval is added to these scalar bars. Ordered steps and within-run observations do not become independent replications.

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Disjoint field-of-view extension

To test whether communication is necessary (not merely beneficial) when observations are non-overlapping, we extend Study 5 to 3 agents each observing a 2-position disjoint window of a 6-position state space (chance-level accuracy 0.167).

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Disjoint field-of-view... (part 2)

Isolated agents achieve mean accuracy 0.35 — above chance but far from decisive, since no single agent can infer the global state from a partial window alone. Communicating agents pool complementary beliefs to reach 0.55 (gap 0.19).

Disjoint field-of-view… (part 3)

This is now a powered result, not an illustrative point estimate. Across 128 independent seeds the isolated accuracy is 0.326 (95% CI 0.320–0.332) and the communicating accuracy is 0.493 (95% CI 0.487–0.499), both clearing the 0.167 chance baseline — isolated agents are *not* at chance,

Disjoint field-of-view... (part 4)

since a partial FOV plus majority voting still carries some signal.

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Disjoint field-of-view… (part 5)

The paired Wilcoxon signed-rank test (communicating vs. isolated, matched by seed) gives $p = 9.35 \times 10^{-23}$, effect size $r = 1.000$ (large): communicating beats isolated on every one of the 128 seeds. This agreement saturates the rank-biserial effect at its upper bound;

Disjoint field-of-view... (part 6)

it does not turn the reported Wilcoxon signed-rank p-value into an exact paired sign-test probability. The paired mean gap and its interval quantify the accuracy contrast in its native units.

Disjoint field-of-view… (part 7)

Given isolated performance is above chance, the precise claim is not that communication is *logically* necessary for any signal at all, but that it is necessary to approach the communicating-level accuracy under fully disjoint observations: the gap between the two conditions is significant, large,

Disjoint field-of-view… (part 8)

and reproducible, unlike the binary-complement contrast above.

Disjoint field-of-view... (part 9)

We separately quantify EFE-guided navigation rather than asserting an unquantified "widens the gap" effect. In a matched but smaller-scale disjoint-FOV movement-policy comparison (2 agents, 4-position binary-state grid, belief sharing active in both arms),

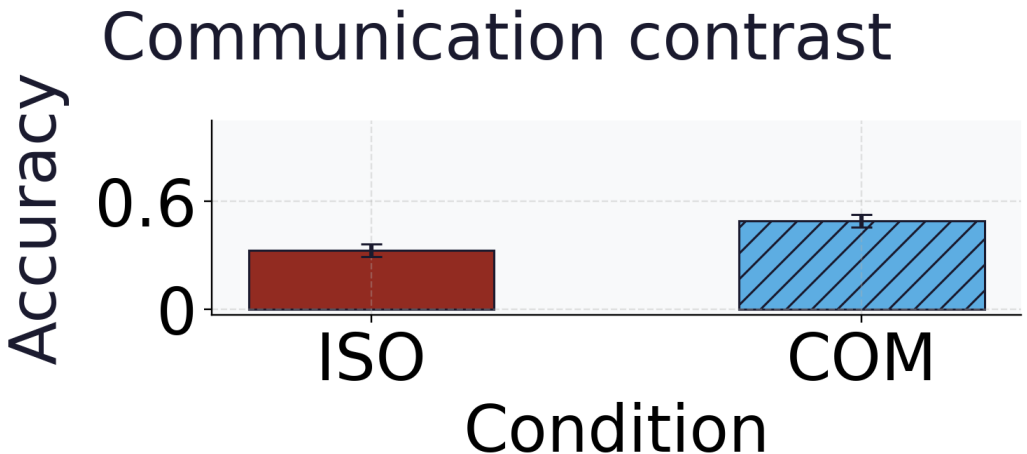
Disjoint field-of-view... (part 10)

EFE-guided accuracy is 0.975 versus 0.977 for random movement ($p = 0.1046$, negligible effect). This comparison did not resolve an accuracy difference in the configured near-ceiling regime with belief sharing active; it does not establish equivalence of the movement policies.

Disjoint field-of-view… (part 11)

We report this as the null result it is rather than claiming an unmeasured EFE benefit. fig. 30 summarizes the necessity result.

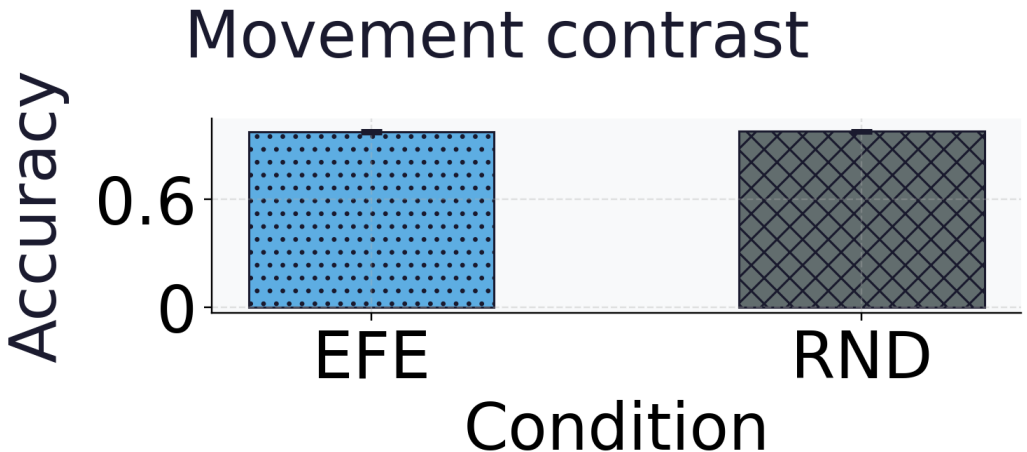
Disjoint field-of-view... (part 12)



Disjoint field-of-view… (part 13)

ISO mean 0.3262, SD 0.03472; COM mean 0.4928, SD 0.0349. COM−ISO = +0.1666.

Disjoint field-of-view... (part 14)



Disjoint field-of-view… (part 15)

EFE mean 0.9749, SD 0.01134; RND mean 0.9768, SD 0.01019. EFE−RND = -0.001953.

Disjoint field-of-view… (part 16)

ISO: isolated; COM: communicating; EFE: EFE-guided; RND: random movement. Hatches preserve condition identity.

Disjoint field-of-view… (part 17)

Whiskers show standard deviations across configured seeds, not confidence intervals. Accuracy is a fraction correct.

Disjoint field-of-view… (part 18)

Separate communication and movement-policy contrasts in the declared disjoint-view worlds. A near-ceiling movement contrast does not imply a general navigation advantage.

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